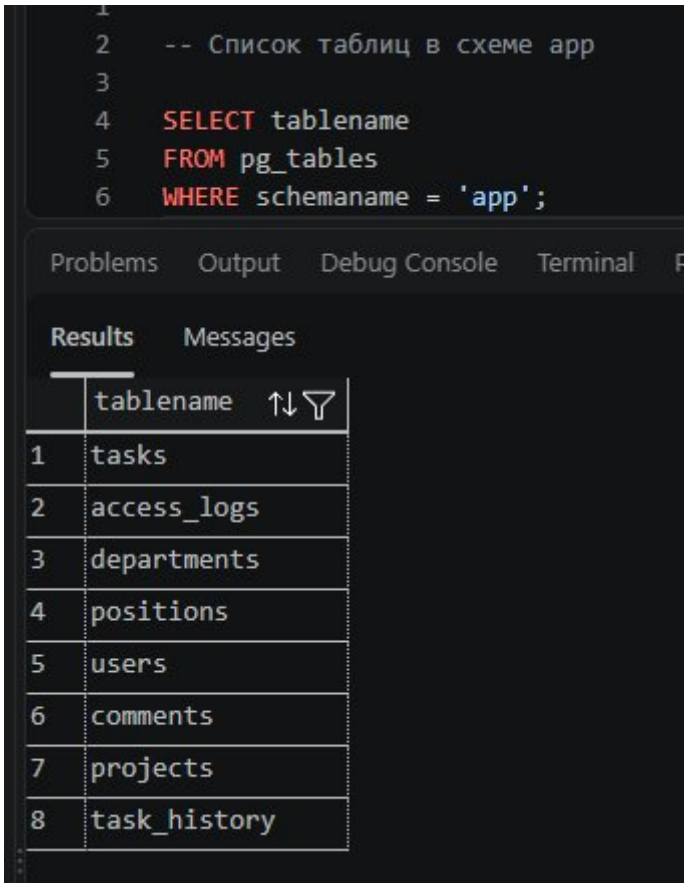
Практическая работа №6

1.1



```
2   -- Список таблиц в схеме app
3
4   SELECT tablename
5   FROM pg_tables
6   WHERE schemaname = 'app';
```

Problems Output Debug Console Terminal

Results Messages

	tablename
1	tasks
2	access_logs
3	departments
4	positions
5	users
6	comments
7	projects
8	task_history

localhost | corporate_tasks | schema: session default

```sql
-- Проверка данных

SELECT 'departments' AS table_name, COUNT(*)
FROM app.departments

UNION ALL

SELECT 'positions', COUNT(*)
FROM app.positions

UNION ALL

SELECT 'users', COUNT(*)
FROM app.users

UNION ALL

SELECT 'projects', COUNT(*)
FROM app.projects

UNION ALL

SELECT 'tasks', COUNT(*)
FROM app.tasks

UNION ALL

SELECT 'comments', COUNT(*)
FROM app.comments;
```

Problems Output Debug Console Terminal Ports PostgreSQL Qu

Results Messages

	table_name	count
1	departments	3
2	positions	5
3	users	5
4	projects	4
5	tasks	14
6	comments	6

```sql

SELECT grantee, table_name, privilege_type
FROM information_schema.role_table_grants
WHERE grantee IN (
    'app_user',
    'app_manager',
    'app_admin'
)
ORDER BY grantee, table_name;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query R

Results Messages

grantee	table_name	privilege…

```sql
-- Связь задач с пользователями
SELECT
    t.task_id,
    t.title,
    t.status,
    u.username AS assignee,
    p.name AS project
FROM app.tasks t
JOIN app.users u ON t.assignee_id = u.user_id
JOIN app.projects p ON t.project_id = p.project_id
LIMIT 10;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages localh

	task_id	title	status	assignee	project
1	2	Implement navigation	todo	charlie	Website Redesign
2	3	Setup CI/CD	todo	charlie	Website Redesign
3	4	Create color palette	review	bob	Website Redesign
4	5	Prepare responsive layout	todo	bob	Website Redesign
5	6	Test website	todo	charlie	Website Redesign
6	7	Requirements analysis	in_progress	alice	Mobile App
7	8	Architecture design	todo	charlie	Mobile App
8	9	Setup project structure	todo	bob	Mobile App
9	10	Design login screen	todo	bob	Mobile App
10	11	Implement API	todo	charlie	Mobile App

2.1

```
1
2   -- 1. Включаем RLS для таблицы задач
3
4   ALTER TABLE app.tasks ENABLE ROW LEVEL SECURITY;
5
6   -- 2. Проверяем
7
8   SELECT relname, relrowsecurity
9   FROM pg_class
10  WHERE relname = 'tasks';
11
```

Problems Output Debug Console Terminal Ports PostgreSQL Qu

Results Messages

	relname	relrowsecurity
1	tasks	True

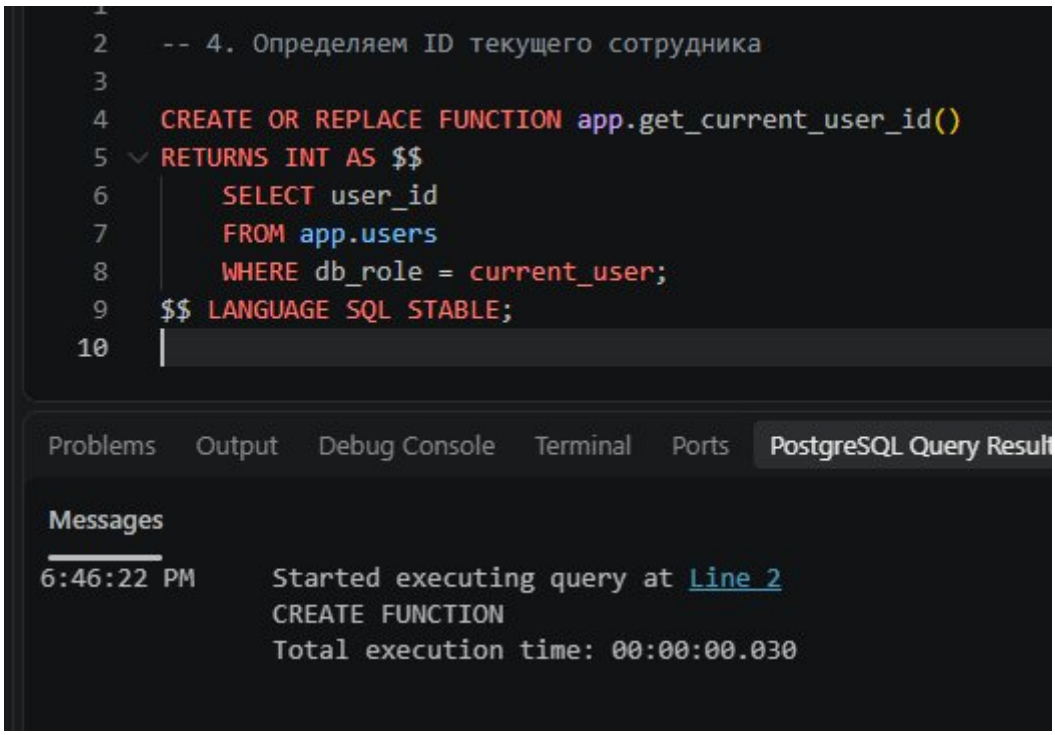
\

```
1
2   -- 3. Проверка RLS без политик
3
4   SET ROLE app_user;
5
6   SELECT COUNT(*) FROM app.tasks;
7
8   RESET ROLE;
9
```

Problems Output Debug Console Terminal Ports

Results Messages

	count
1	0


-- 4. Определяем ID текущего сотрудника
CREATE OR REPLACE FUNCTION app.get_current_user_id()
RETURNS INT AS $$
SELECT user_id
FROM app.users
WHERE db_role = current_user;
$$ LANGUAGE SQL STABLE;
Problems
Output
Debug Console
Terminal
Ports
PostgreSQL Query Result
Messages
6:46:22 PM
Started executing query at Line 2
CREATE FUNCTION
Total execution time: 00:00:00.030

```sql
-- 5. Пользователь видит только свои задачи

CREATE POLICY tasks_select_own ON app.tasks
    FOR SELECT
    TO app_user
    USING (
        assignee_id = app.get_current_user_id()
    );

-- 6. Может создавать задачи только для себя

CREATE POLICY tasks_insert_own ON app.tasks
    FOR INSERT
    TO app_user
    WITH CHECK (
        assignee_id = app.get_current_user_id()
    );

-- 7. Может изменять только свои задачи

CREATE POLICY tasks_update_own ON app.tasks
    FOR UPDATE
    TO app_user
    USING (
        assignee_id = app.get_current_user_id()
    )
    WITH CHECK (
        assignee_id = app.get_current_user_id()
    );

-- 8. Политика удаления своих задач

CREATE POLICY tasks_delete_own ON app.tasks
    FOR DELETE
    TO app_user
    USING (
        assignee_id = app.get_current_user_id()
    );
```

```sql
--dev_alice
SELECT task_id, title, project_id, assignee_id
FROM app.tasks;
```

Problems Output Debug Console Terminal Ports **PostgreSQL Query Results**

Results Messages

	task_id	title	project_id	assignee_id
1	7	Requirements analysis	2	1
2	12	Test mobile app	2	1
3	14	Admin Test Task	1	1

2.2

```sql
-- Проверка, является ли пользователь владельцем проекта

CREATE OR REPLACE FUNCTION app.is_project_manager(p_project_id INT)
RETURNS BOOLEAN AS $$
    SELECT EXISTS (
        SELECT 1
        FROM app.projects p
        WHERE p.project_id = $1
          AND p.owner_id = app.get_current_user_id()
    );
$$ LANGUAGE SQL STABLE;
```

Problems Output Debug Console Terminal Ports **PostgreSQL Query Results**

Messages

```
7:02:22 PM     Started executing query at Line 2
               CREATE FUNCTION
               Total execution time: 00:00:00.004
```

```sql
-- 1. Менеджер видит свои задачи
-- или задачи в проектах, которыми руководит

CREATE POLICY tasks_select_managed ON app.tasks
    FOR SELECT
    TO app_manager
    USING (
        assignee_id = app.get_current_user_id()
        OR
        app.is_project_manager(project_id)
    );

-- 2. Менеджер может создавать задачи
-- для себя или в своих проектах
CREATE POLICY tasks_insert_managed ON app.tasks
    FOR INSERT
    TO app_manager
    WITH CHECK (
        assignee_id = app.get_current_user_id()
        OR
        app.is_project_manager(project_id)
    );

-- 3. Менеджер может изменять задачи
-- для себя или в своих проектах
CREATE POLICY tasks_update_managed ON app.tasks
    FOR UPDATE
    TO app_manager
    USING (
        assignee_id = app.get_current_user_id()
        OR
        app.is_project_manager(project_id)
    )
    WITH CHECK (
        assignee_id = app.get_current_user_id()
        OR
        app.is_project_manager(project_id)
    );

-- 4. Политика удаления задач
CREATE POLICY tasks_delete_managed ON app.tasks
    FOR DELETE
    TO app_manager
    USING (
        assignee_id = app.get_current_user_id()
        OR
        app.is_project_manager(project_id)
    );

```

localhost | corporate_tasks | schema: session default

```
1  --pm_bob
2  SELECT task_id, title, project_id, assignee_id
3  FROM app.tasks;
```

Problems Output Debug Console Terminal Ports **PostgreSQL Query Results**

Results Messages

	task_id	title	project_id	assignee_id
1	4	Create color palette	1	2
2	5	Prepare responsive layout	1	2
3	9	Setup project structure	2	2
4	10	Design login screen	2	2
5	1	Design homepage	1	2

2.3

```
2  -- Задание 2.3
3  -- Полный доступ администратора к задачам
4
5  CREATE POLICY tasks_admin_all ON app.tasks
6      FOR ALL
7      TO app_admin
8      USING (true)
9      WITH CHECK (true);
10
```

```
1  -- admin_diana
2  SELECT task_id, title, project_id, assignee_id
3  FROM app.tasks
4  ORDER BY task_id;
```

Problems Output Debug Console Terminal Ports **PostgreSQL Query Results**

Results Messages

	task_id	title	project_id	assignee_id
1	1	Design homepage	1	2
2	2	Implement navigation	1	3
3	3	Setup CI/CD	1	3
4	4	Create color palette	1	2
5	5	Prepare responsive layout	1	2
6	6	Test website	1	3
7	7	Requirements analysis	2	1
8	8	Architecture design	2	3
9	9	Setup project structure	2	2
10	10	Design login screen	2	2
11	11	Implement API	2	3
12	12	Test mobile app	2	1
13	13	Manager test task	4	*NULL*
14	14	Admin Test Task	1	1

3.1

```
1  -- Задание 3.1
2  -- Включаем RLS для комментариев
3
4  ALTER TABLE app.comments ENABLE ROW LEVEL SECURITY;
```

```sql
-- 1. Чтение комментариев
CREATE POLICY comments_user_read ON app.comments
    FOR SELECT
    TO app_user
    USING (
        user_id = app.get_current_user_id()
        OR
        (
            task_id IN (
                SELECT task_id
                FROM app.tasks
                WHERE assignee_id = app.get_current_user_id()
            )
            AND is_internal = false
        )
    );
-- 2. Создание комментариев
CREATE POLICY comments_user_insert ON app.comments
    FOR INSERT
    TO app_user
    WITH CHECK (
        task_id IN (
            SELECT task_id
            FROM app.tasks
            WHERE assignee_id = app.get_current_user_id()
        )
        AND user_id = app.get_current_user_id()
        AND is_internal = false
    );
-- 3. Изменение комментариев
CREATE POLICY comments_user_update ON app.comments
    FOR UPDATE
    TO app_user
    USING (
        user_id = app.get_current_user_id()
    )
    WITH CHECK (
        user_id = app.get_current_user_id()
        AND is_internal = false
    );
-- 4. Удаление комментариев
CREATE POLICY comments_user_delete ON app.comments
    FOR DELETE
    TO app_user
    USING (
        user_id = app.get_current_user_id()
    );
```

```sql

-- Менеджер читает все комментарии
CREATE POLICY comments_manager_read ON app.comments
    FOR SELECT
    TO app_manager
    USING (true);

-- Менеджер создаёт комментарии
CREATE POLICY comments_manager_insert ON app.comments
    FOR INSERT
    TO app_manager
    WITH CHECK (true);

-- Менеджер изменяет комментарии
CREATE POLICY comments_manager_update ON app.comments
    FOR UPDATE
    TO app_manager
    USING (true)
    WITH CHECK (true);

-- Менеджер удаляет комментарии
CREATE POLICY comments_manager_delete ON app.comments
    FOR DELETE
    TO app_manager
    USING (true);
```

```sql

 -- Полный доступ администратора

 CREATE POLICY comments_admin_all ON app.comments
    FOR ALL
    TO app_admin
    USING (true)
    WITH CHECK (true);

```

4.1

```sql
    -- Включение RLS для проектов

    ALTER TABLE app.projects ENABLE ROW LEVEL SECURITY;
```

```sql
-- Обычный пользователь видит проекты,
-- в которых у него есть задачи,
-- или проекты, которыми он владеет

CREATE POLICY projects_user_read ON app.projects
    FOR SELECT
    TO app_user
    USING (
        project_id IN (
            SELECT project_id
            FROM app.tasks
            WHERE assignee_id = app.get_current_user_id()
        )
        OR
        owner_id = app.get_current_user_id()
    );
```

```sql

-- Менеджер видит свои проекты
-- и проекты своего отдела

CREATE POLICY projects_manager_read ON app.projects
    FOR SELECT
    TO app_manager
    USING (
        owner_id = app.get_current_user_id()
        OR
        department_id = (
            SELECT department_id
            FROM app.users
            WHERE user_id = app.get_current_user_id()
        )
    );

-- Менеджер может управлять проектами,
-- владельцем которых является

CREATE POLICY projects_manager_write ON app.projects
    FOR ALL
    TO app_manager
    USING (
        owner_id = app.get_current_user_id()
    )
    WITH CHECK (
        owner_id = app.get_current_user_id()
    );
```

```sql
--dev_alice
SELECT
    project_id,
    name,
    owner_id,
    department_id
FROM app.projects
ORDER BY project_id;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	project_id	name	owner_id	department_id
1	1	Website Redesign	1	1
2	2	Mobile App	1	1
3	3	Test Project	1	1
4	4	Test Project	1	1

```sql
--pm_bob

SELECT
    project_id,
    name,
    owner_id,
    department_id
FROM app.projects
ORDER BY project_id;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	project_id	name	owner_id	department_id
1	1	Website Redesign	1	1
2	2	Mobile App	1	1
3	3	Test Project	1	1
4	4	Test Project	1	1

```sql
-- admin_diana
SELECT
    project_id,
    name,
    owner_id,
    department_id
FROM app.projects
ORDER BY project_id;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	project_id	name	owner_id	department_id
1	1	Website Redesign	1	1
2	2	Mobile App	1	1
3	3	Test Project	1	1
4	4	Test Project	1	1

4.2

```sql
-- Включение RLS для users

ALTER TABLE app.users ENABLE ROW LEVEL SECURITY;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Messages

```
7:48:03 PM     Started executing query at Line 1
               ALTER TABLE
               Total execution time: 00:00:00
```

```sql

-- Обычный пользователь видит только себя

CREATE POLICY users_self_read ON app.users
    FOR SELECT
    TO app_user
    USING (
        user_id = app.get_current_user_id()
    );

-- Обычный пользователь может изменять
-- только свою запись, если у него есть право UPDATE

CREATE POLICY users_self_update ON app.users
    FOR UPDATE
    TO app_user
    USING (
        user_id = app.get_current_user_id()
    )
    WITH CHECK (
        user_id = app.get_current_user_id()
    );
```

```sql
-- app_manager: видит пользователей своего отдела
CREATE POLICY users_department_read ON app.users
    FOR SELECT
    TO app_manager
    USING (
        department_id = (
            SELECT department_id FROM app.users
            WHERE user_id = app.get_current_user_id()
        )
    );
```

localhost | corporate_tasks | schema: session default

```sql
-- app_admin: полный доступ
CREATE POLICY users_admin_all ON app.users
    FOR ALL
    TO app_admin
    USING (true)
    WITH CHECK (true);
```

```sql
--dev_alice
SELECT
    user_id,
    username,
    full_name,
    department_id
FROM app.users
ORDER BY user_id;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	user_id	username	full_name	department_id
1	1	alice	Alice Johnson	1

```sql
--pm_bob
SELECT
    user_id,
    username,
    full_name,
    department_id
FROM app.users
ORDER BY user_id;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	user_id	username	full_name	department_id
1	1	alice	Alice Johnson	1
2	2	bob	Bob Smith	1
3	3	charlie	Charlie Brown	1

4.3

```
1  SELECT
2      schemaname,
3      tablename,
4      rowsecurity
5  FROM pg_tables
6  WHERE schemaname = 'app'
7  ORDER BY tablename;
```

Problems Output Debug Console Terminal Ports PostgreSQL Qu

Results Messages

	schemaname	tablename	rowsecurity
1	app	access_logs	False
2	app	comments	True
3	app	departments	False
4	app	positions	False
5	app	projects	True
6	app	task_history	False
7	app	tasks	True
8	app	users	True

5.1

```
1  --dev_alice
2  SELECT task_id, ti
3  FROM app.tasks;
```

Problems Output Debug Co

Results Messages

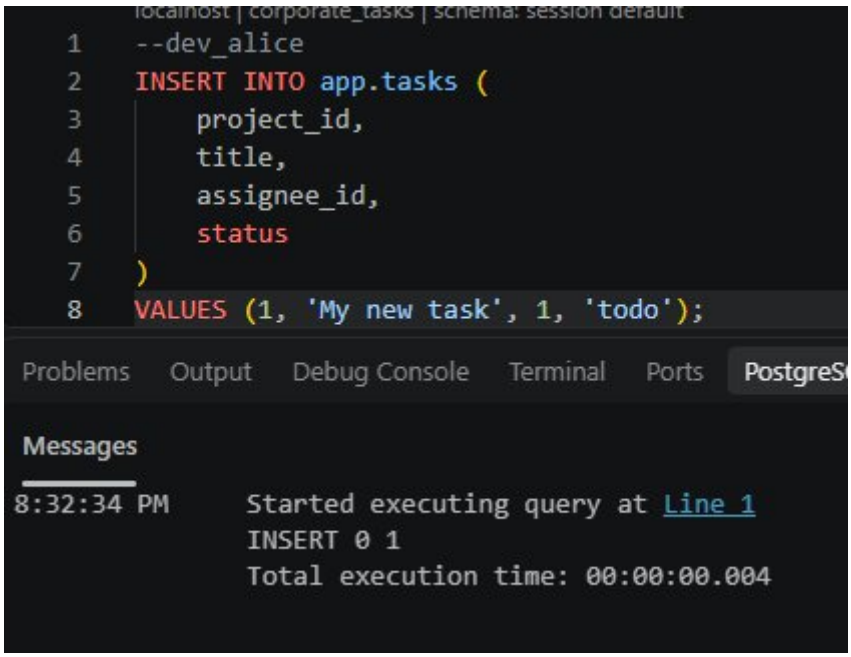
	task_id	title
1	7	Require
2	12	Test mo
3	14	Admin T

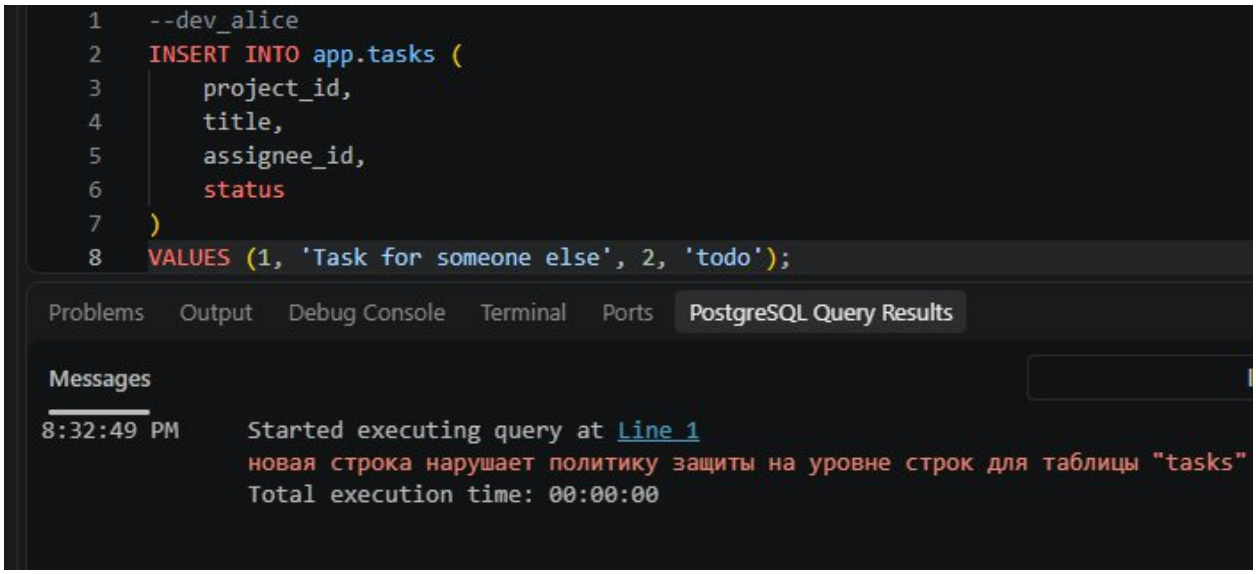
```
1  --dev_alice
2  SELECT COUNT(*) FROM app.tasks;
3
```

Problems Output Debug Console Terminal Por

Results Messages

	count
1	3


localhost | corporate_tasks | schema: session default
1 --dev_alice
2 INSERT INTO app.tasks (
3 project_id,
4 title,
5 assignee_id,
6 status
7)
8 VALUES (1, 'My new task', 1, 'todo');
Problems Output Debug Console Terminal Ports PostgreS
Messages
8:32:34 PM Started executing query at Line 1
INSERT 0 1
Total execution time: 00:00:00.004

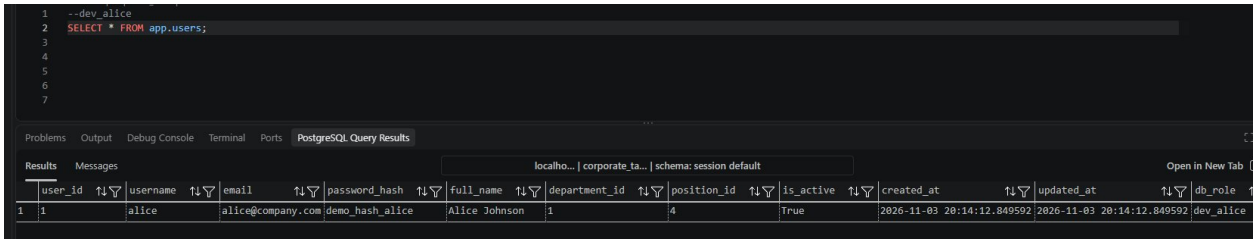

1 --dev_alice
2 INSERT INTO app.tasks (
3 project_id,
4 title,
5 assignee_id,
6 status
7)
8 VALUES (1, 'Task for someone else', 2, 'todo');
Problems Output Debug Console Terminal Ports PostgreSQL Query Results
Messages
8:32:49 PM Started executing query at Line 1
новая строка нарушает политику защиты на уровне строк для таблицы "tasks"
Total execution time: 00:00:00

```
1  --dev_alice
2  UPDATE app.tasks
3  SET status = 'in_progress'
4  WHERE task_id = 12;
5
6
7
8
```

Problems Output Debug Console Terminal Ports Postgre

Messages

:33:22 PM Started executing query at Line 1
UPDATE 1
Total execution time: 00:00:00




```
1  --dev_alice
2  SELECT comment_id, task_id, user_id, is_internal
3  FROM app.comments;
4
5
6
7
8
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	comment_id	task_id	user_id	is_internal
1	1	1	1	False
2	3	7	1	False
3	5	1	1	False
4	6	1	1	False

```sql
--dev_alice
INSERT INTO app.comments (
    task_id,
    user_id,
    content,
    is_internal
)
VALUES (7, 1, 'My RLS test comment', false);
```

Problems Output Debug Console Terminal Ports PostgreSQL

Messages

```
8:34:10 PM     Started executing query at Line 1
               INSERT 0 1
               Total execution time: 00:00:00.001
```

localhost | corporate_tasks | schema: session default

```sql
--dev_alice
INSERT INTO app.comments (
    task_id,
    user_id,
    content,
    is_internal
)
VALUES (7, 1, 'My RLS test comment', true);
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Messages

```
8:34:35 PM     Started executing query at Line 1
               новая строка нарушает политику защиты на уровне строк для таблицы "comments"
               Total execution time: 00:00:00
```

5.2

```sql
--pm_bob
-- Тест 1. Задачи менеджера
SELECT task_id, title, project_id, assignee_id
FROM app.tasks;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	task_id	title	project_id	assignee_id
1	4	Create color palette	1	2
2	5	Prepare responsive layout	1	2
3	9	Setup project structure	2	2
4	10	Design login screen	2	2
5	1	Design homepage	1	2

```sql
--pm_bob
-- Тест 2. Чтение внутренних комментариев
SELECT comment_id, is_internal
FROM app.comments;
```

Problems Output Debug Console Terminal Ports Postgre

Results Messages

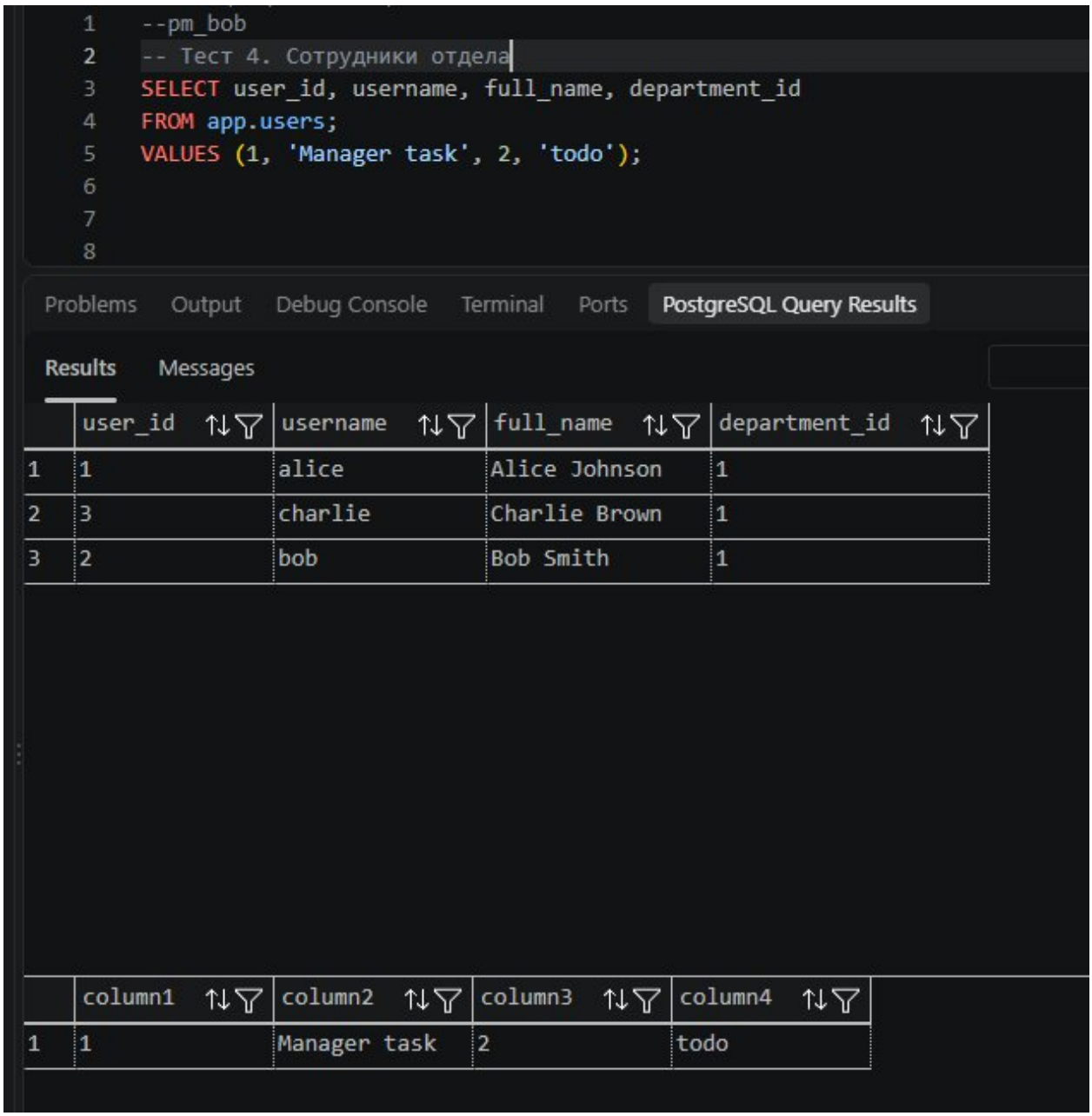
	comment_id	is_internal
1	1	False
2	2	False
3	3	False
4	4	True
5	5	False
6	6	False
7	7	False

```sql
--pm_bob
-- Тест 3. Создание задачи менеджером
INSERT INTO app.tasks (
    project_id,
    title,
    assignee_id,
    status
)
```

Problems Output Debug Console Terminal Ports PostgreSQL Query

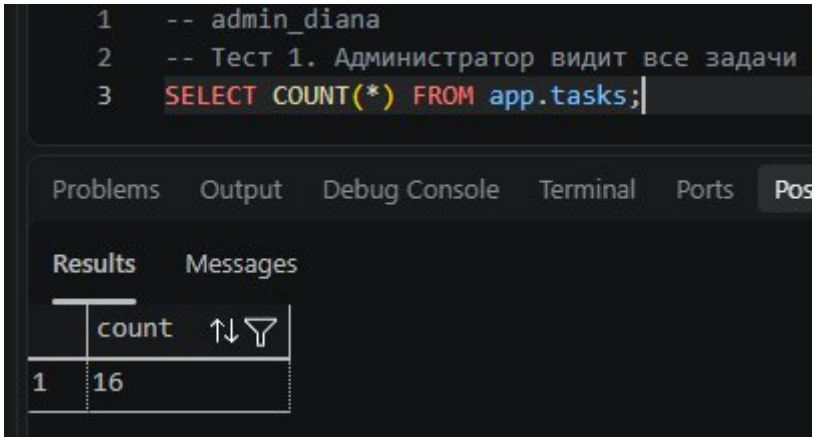
Messages

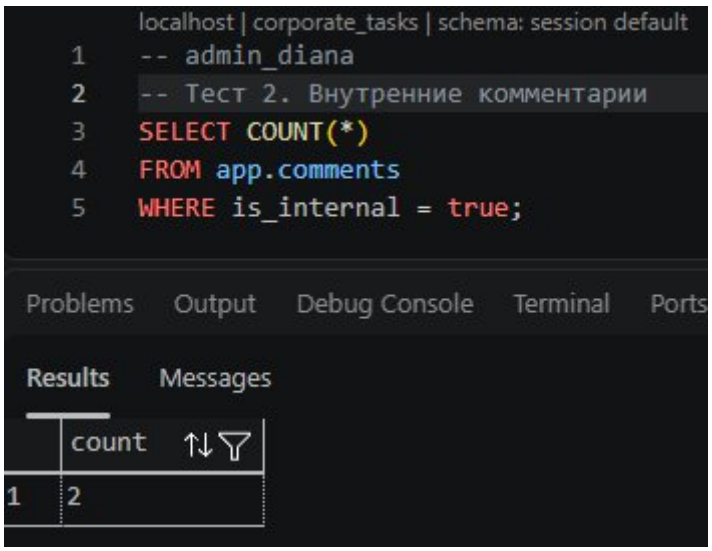
```
8:36:51 PM    Started executing query at Line 1
              INSERT 0 1
              Total execution time: 00:00:00.002
```

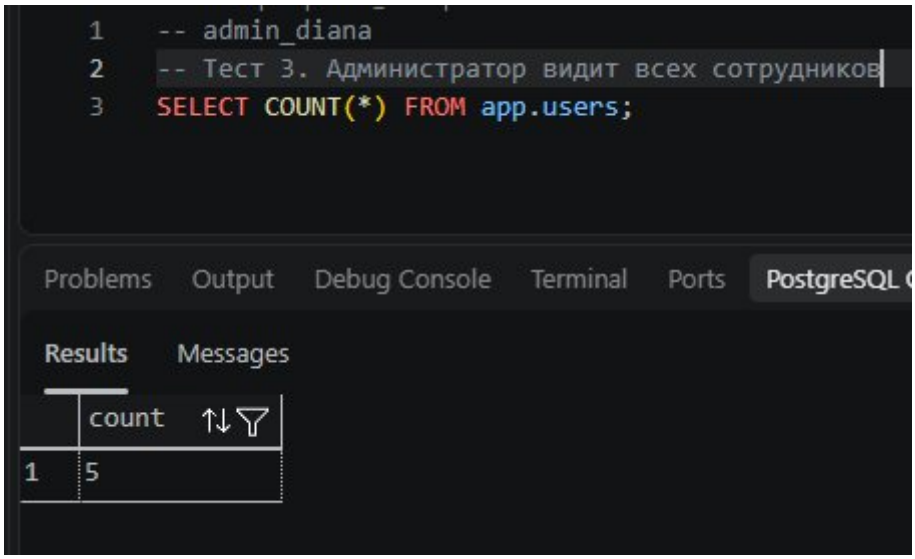

1 --pm_bob
2 -- Тест 4. Сотрудники отдела
3 SELECT user_id, username, full_name, department_id
4 FROM app.users;
5 VALUES (1, 'Manager task', 2, 'todo');
6
7
8
Problems Output Debug Console Terminal Ports PostgreSQL Query Results
Results Messages
user_id username full_name department_id
1 1 alice Alice Johnson 1
2 3 charlie Charlie Brown 1
3 2 bob Bob Smith 1
column1 column2 column3 column4
1 1 Manager task 2 todo


localhost | corporate_tasks | schema: session default
--pm_bob
-- Тест 5. Внутренний комментарий
INSERT INTO app.comments (
task_id,
user_id,
content,
is_internal
)
VALUES (
1,
2,
'Manager internal comment',
true
);
Problems
Output
Debug Console
Terminal
Ports
PostgreSQL
Messages
8:37:39 PM
Started executing query at Line 1
INSERT 0 1
Total execution time: 00:00:00.001

5.3


-- admin_diana
-- Тест 1. Администратор видит все задачи
SELECT COUNT(*) FROM app.tasks;
Problems
Output
Debug Console
Terminal
Ports
Pos
Results
Messages
count
1
16


localhost | corporate_tasks | schema: session default
1 -- admin_diana
2 -- Тест 2. Внутренние комментарии
3 SELECT COUNT(*)
4 FROM app.comments
5 WHERE is_internal = true;
Problems Output Debug Console Terminal Ports
Results Messages
count
1 2


1 -- admin_diana
2 -- Тест 3. Администратор видит всех сотрудников
3 SELECT COUNT(*) FROM app.users;
Problems Output Debug Console Terminal Ports PostgreSQL
Results Messages
count
1 5

5.4

```
1  --dev_alice
2  -- План выполнения запроса от имени dev_alice
3  EXPLAIN (ANALYZE, BUFFERS, VERBOSE)
4  SELECT * FROM app.tasks;
5
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Results

Results Messages

	QUERY PLAN
1	Seq Scan on app.tasks (cost=0.00..44.12 rows=1 wi…
2	Output: task_id, project_id, title, description,…
3	Filter: (tasks.assignee_id = app.get_current_use…
4	Rows Removed by Filter: 12
5	Buffers: shared hit=17
6	Planning:
7	Buffers: shared hit=1
8	Planning Time: 0.222 ms
9	Execution Time: 0.103 ms

```
1  -- admin_diana
2  -- План выполнения запроса администратора
3  EXPLAIN (ANALYZE, BUFFERS, VERBOSE)
4  SELECT * FROM app.tasks;;
```

Problems Output Debug Console Terminal Ports PostgreSQL Qu

Results Messages

	QUERY PLAN
1	Seq Scan on app.tasks (cost=0.00..11.30 rows=130 …
2	Output: task_id, project_id, title, description,…
3	Buffers: shared hit=1
4	Planning Time: 0.170 ms
5	Execution Time: 0.037 ms

```sql
--dev_alice
-- Проверка поиска задач по assignee_id
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM app.tasks
WHERE assignee_id = app.get_current_user_id();
```

Problems Output Debug Console Terminal Ports PostgreSQL Query R

Results Messages

	QUERY PLAN
1	Seq Scan on tasks (cost=0.00..44.12 rows=1 width=…
2	Filter: (assignee_id = app.get_current_user_id())
3	Rows Removed by Filter: 12
4	Buffers: shared hit=17
5	Planning:
6	Buffers: shared hit=1
7	Planning Time: 0.146 ms
8	Execution Time: 0.093 ms

```sql
-- Индекс для ускорения поиска задач по исполнителю

CREATE INDEX IF NOT EXISTS idx_tasks_assignee
ON app.tasks (assignee_id);
```

Problems Output Debug Console Terminal Ports PostgreSQL Query

Messages

```
8:47:25 PM     Started executing query at Line 2
               CREATE INDEX
               Total execution time: 00:00:00.039
```

```sql
--dev_alice
-- Проверка после создания индекса
EXPLAIN (ANALYZE, BUFFERS)
SELECT *
FROM app.tasks
WHERE assignee_id = app.get_current_user_id();
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Result

Results Messages

	QUERY PLAN
1	Seq Scan on tasks (cost=0.00..5.20 rows=1 width=5…
2	Filter: (assignee_id = app.get_current_user_id())
3	Rows Removed by Filter: 12
4	Buffers: shared hit=17
5	Planning:
6	Buffers: shared hit=17 read=1
7	Planning Time: 1.907 ms
8	Execution Time: 0.098 ms

6.1

```sql

-- 1. Все таблицы с RLS

SELECT
    schemaname,
    tablename,
    rowsecurity AS "RLS включён"
FROM pg_tables
WHERE schemaname = 'app'
ORDER BY tablename;

```

Problems Output Debug Console Terminal Ports PostgreSQL Query Resu

Results Messages

	schemaname	tablename	RLS включён
1	app	access_logs	False
2	app	comments	True
3	app	departments	False
4	app	positions	False
5	app	projects	True
6	app	task_history	False
7	app	tasks	True
8	app	users	True

```sql
-- 2. Все политики

SELECT
    tablename,
    policyname,
    cmd AS "Команда",
    roles AS "Роли",
    qual AS "USING условие",
    with_check AS "CHECK условие"
FROM pg_policies
WHERE schemaname = 'app'
ORDER BY tablename, policyname;
```

	tablename	policyname	Команда	Роли	USING условие	CHECK условие
1	comments	comments_admin_all	ALL	{app_admin}	true	true
2	comments	comments_manager_delete	DELETE	{app_manager}	true	*NULL*
3	comments	comments_manager_insert	INSERT	{app_manager}	*NULL*	true
4	comments	comments_manager_read	SELECT	{app_manager}	true	*NULL*
5	comments	comments_manager_update	UPDATE	{app_manager}	true	true
6	comments	comments_user_delete	DELETE	{app_user}	(user_id = app.get_current_user_id())	*NULL*
7	comments	comments_user_insert	INSERT	{app_user}	*NULL*	((task_id IN (SELECT tasks.task_id FROM app.ta…
8	comments	comments_user_read	SELECT	{app_user}	((user_id = app.get_current_user_id()) OR ((task_i…	*NULL*
9	comments	comments_user_update	UPDATE	{app_user}	(user_id = app.get_current_user_id())	((user_id = app.get_current_user_id()) AND (is_int…
10	projects	projects_admin_all	ALL	{app_admin}	true	true
11	projects	projects_manager_read	SELECT	{app_manager}	((owner_id = app.get_current_user_id()) OR (depart…	*NULL*
12	projects	projects_manager_write	ALL	{app_manager}	(owner_id = app.get_current_user_id())	(owner_id = app.get_current_user_id())
13	projects	projects_user_read	SELECT	{app_user}	(app.user_has_task_in_project(project_id, app.get_…	*NULL*
14	tasks	tasks_admin_all	ALL	{app_admin}	true	true
15	tasks	tasks_delete_managed	DELETE	{app_manager}	((assignee_id = app.get_current_user_id()) OR app.…	*NULL*
16	tasks	tasks_delete_own	DELETE	{app_user}	(assignee_id = app.get_current_user_id())	*NULL*
17	tasks	tasks_insert_managed	INSERT	{app_manager}	*NULL*	((assignee_id = app.get_current_user_id()) OR app.…
18	tasks	tasks_insert_own	INSERT	{app_user}	*NULL*	(assignee_id = app.get_current_user_id())
19	tasks	tasks_select_managed	SELECT	{app_manager}	((assignee_id = app.get_current_user_id()) OR app.…	*NULL*
20	tasks	tasks_select_own	SELECT	{app_user}	(assignee_id = app.get_current_user_id())	*NULL*
21	tasks	tasks_update_managed	UPDATE	{app_manager}	((assignee_id = app.get_current_user_id()) OR app.…	((assignee_id = app.get_current_user_id()) OR app.…
22	tasks	tasks_update_own	UPDATE	{app_user}	(assignee_id = app.get_current_user_id())	(assignee_id = app.get_current_user_id())
23	users	users_admin_all	ALL	{app_admin}	true	true
24	users	users_department_read	SELECT	{app_manager}	(department_id = app.get_current_user_department_i…	*NULL*
25	users	users_self_read	SELECT	{app_user}	(user_id = app.get_current_user_id())	*NULL*
26	users	users_self_update	UPDATE	{app_user}	(user_id = app.get_current_user_id())	(user_id = app.get_current_user_id())

```sql

-- 3. Функции, используемые в политиках

SELECT
    routine_name,
    routine_type,
    data_type
FROM information_schema.routines
WHERE routine_schema = 'app'
  AND (
      routine_name LIKE 'get_%'
      OR routine_name LIKE 'is_%'
      OR routine_name = 'user_has_task_in_project'
  )
ORDER BY routine_name;

```

Problems Output Debug Console Terminal Ports **PostgreSQL Query Results**

Results Messages

	routine_name	routine_type	data_type
1	get_current_user_department_id	FUNCTION	integer
2	get_current_user_id	FUNCTION	integer
3	is_project_manager	FUNCTION	boolean
4	user_has_task_in_project	FUNCTION	boolean

```sql
--dev_alice
-- Количество строк, доступных Алисе
SELECT
    (SELECT COUNT(*) FROM app.tasks) AS tasks,
    (SELECT COUNT(*) FROM app.comments) AS comments,
    (SELECT COUNT(*) FROM app.projects) AS projects,
    (SELECT COUNT(*) FROM app.users) AS users;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query R

Results Messages

	tasks	comments	projects	users
1	4	5	4	1

```sql
--pm_bob
-- Количество строк, доступных Алисе
SELECT
    (SELECT COUNT(*) FROM app.tasks) AS tasks,
    (SELECT COUNT(*) FROM app.comments) AS comments,
    (SELECT COUNT(*) FROM app.projects) AS projects,
    (SELECT COUNT(*) FROM app.users) AS users;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query R

Results Messages

	tasks	comments	projects	users
1	6	8	4	3

```sql
-- admin_diana
-- Количество строк, доступных Алисе
SELECT
    (SELECT COUNT(*) FROM app.tasks) AS tasks,
    (SELECT COUNT(*) FROM app.comments) AS comments,
    (SELECT COUNT(*) FROM app.projects) AS projects,
    (SELECT COUNT(*) FROM app.users) AS users;
```

Problems Output Debug Console Terminal Ports PostgreSQL Query Re

Results Messages

	tasks	comments	projects	users
1	16	8	4	5